

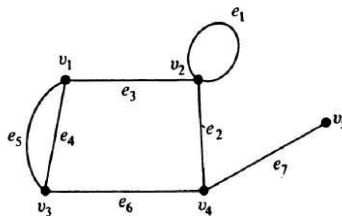
Graph Theory Work-along Sheet

Name: _____; Roll Number: _____

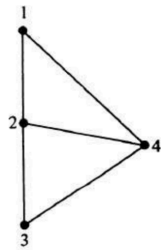
Lecture Batch: _____; Lab Batch: _____

A _____ graph (or simply a _____) $G = (V, E)$ consists of a set of objects $V = \{v_1, v_2, \dots\}$ called _____, and another set $E = \{e_1, e_2, \dots\}$, whose elements are called _____, such that each edge e_k is identified with an _____ pair (v_i, v_j) of vertices. The vertices v_i and v_j associated with edge e_k are called the end vertices of e_k . The most common representation of a graph is by means of a _____, in which the _____ are represented as points and each _____ as a line segment joining its end vertices. Often, this diagram itself is referred to as the _____.

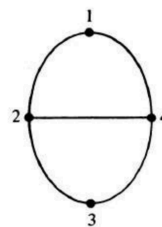
The edge having the same vertex as both its end vertices is called a _____. If more than one edge is associated with a given pair of vertices, then such edges are referred to as _____ edges.



A graph that has neither self-loops nor parallel edges is called a _____ graph.



(a)

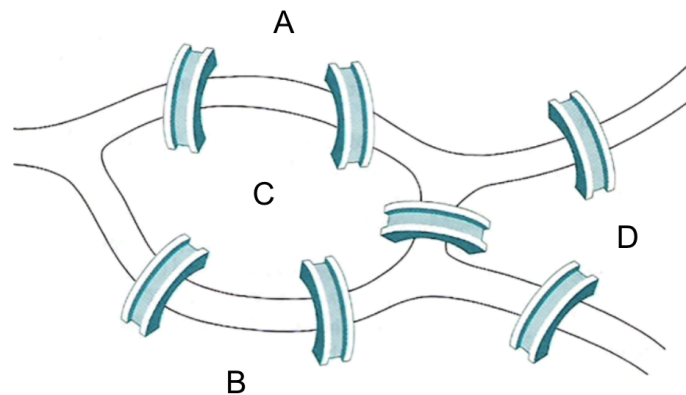


(b)

Question: Are the above two graphs the same or different?

Answer: _____ because incidence between edges and vertices is the same in both cases.

A graph can be used to represent almost any physical situation involving discrete objects and their relationships.



Seven Bridges of Königsberg problem:

- **Location:** The problem is set in Königsberg (now Kaliningrad) on the Pregel River.
- **Geography:** It involves four land areas: two islands (C and D) and two riverbanks (A and B).
- **Connections:** These four land areas are connected by a total of seven bridges.
- **The Challenge:** To find a path that starts at any land area, crosses every bridge exactly once, and returns to the starting point.

Euler's Method: Leonhard Euler modelled the city as a mathematical graph:

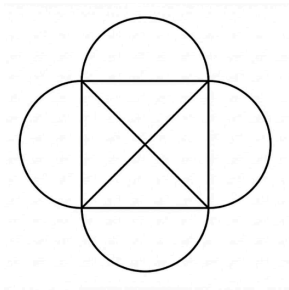
- **Vertices (Points):** Represent the land areas (A, B, C, D).
- **Edges (Lines):** Represent the bridges.

Try to draw the graph of the Königsberg bridge problem.

- **The Solution:** Euler proved that such a path is [possible/impossible] to create. How? Think about it. 🤔

Childhood Activity:

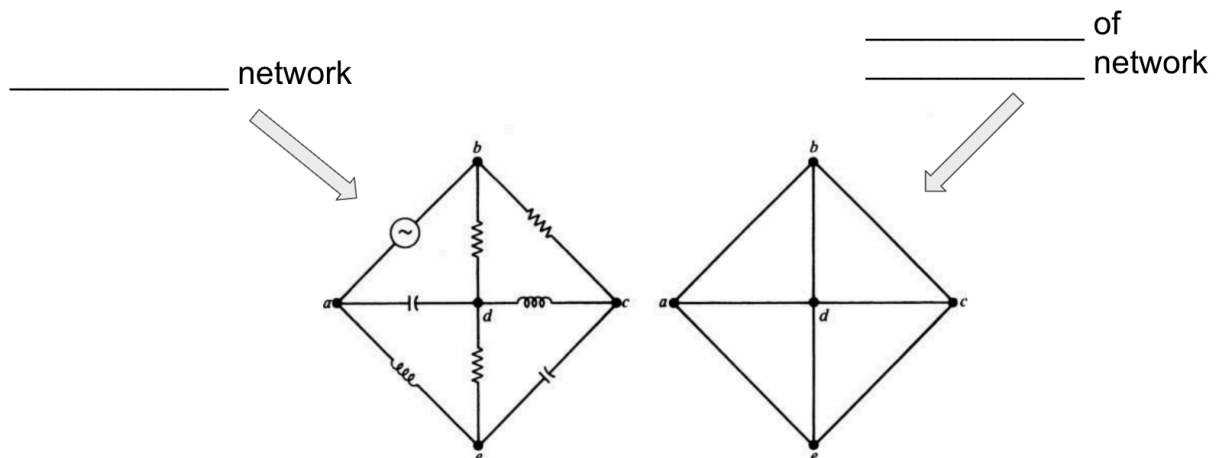
Remember the following childhood activity, where we had to draw the following figure without lifting the pen from the paper and without retracing a line.



[Your first attempt]

[Your second attempt]

Is it possible to do it? [YES / NO]



A graph with a finite number of vertices as well as a finite number of edges is called a _____ graph; otherwise, it is an _____ graph.

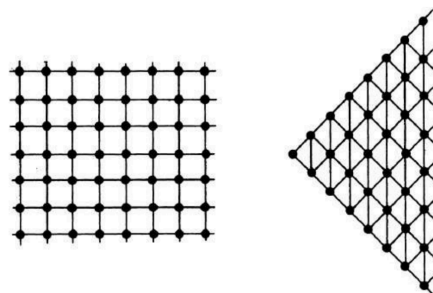
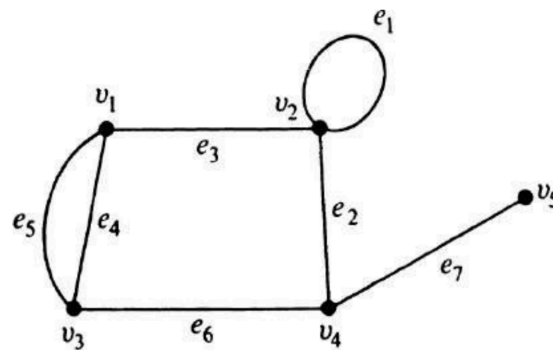


Fig: Portions of two infinite graphs

In this discussion, we shall confine ourselves to the study of finite graphs, and unless otherwise stated, the term “graph” will always mean a finite graph.

The number of edges incident on a vertex v_i , with self-loops counted twice, is called the degree, $d(v_i)$, of vertex v_i . The degree of a vertex is sometimes also referred to as its _____.

Find the degree of all the vertices of the following graph:



$d(v_1) = \underline{\hspace{1cm}}$, $d(v_2) = \underline{\hspace{1cm}}$, $d(v_3) = \underline{\hspace{1cm}}$, $d(v_4) = \underline{\hspace{1cm}}$, $d(v_5) = \underline{\hspace{1cm}}$

Now, add all the degrees.

$$\begin{aligned} & d(v_1) + d(v_2) + d(v_3) + d(v_4) + d(v_5) \\ &= \underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$

Is there any relation between the sum of the degrees of all the vertices and the number of edges? State your observations.

Handshaking Lemma:

The Handshaking Lemma states that the _____ of the degrees of all the vertices in any finite graph must equal twice the number of _____.

This is true because every edge connects exactly two vertices, meaning it contributes exactly 2 to the total sum of the degrees.

$$\sum_{i=1}^N d(v_i) = 2e$$

Now, write the degrees of the vertices of the earlier-mentioned graph in non-increasing order.

_____ $\geq$ _____ $\geq$ _____ $\geq$ _____ $\geq$ _____
 { _____, _____, _____, _____, _____ }

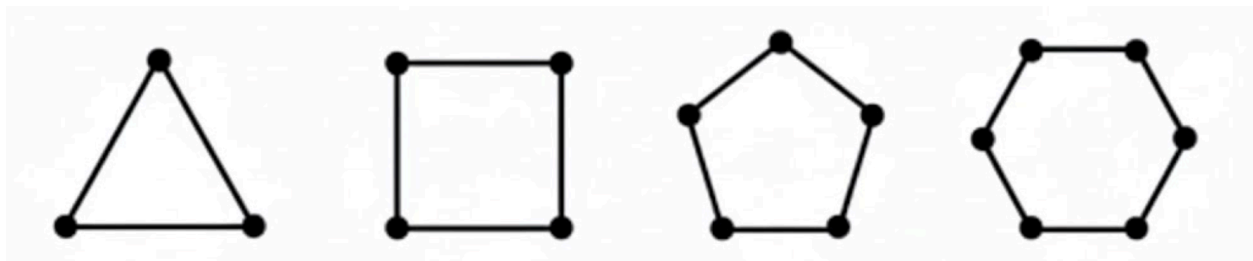
This degree sequence provides a fundamental summary of a graph's structure.

Degree Sequence:

The _____ of a graph is simply a list of the _____ of all its _____ (nodes), typically arranged in _____ order.

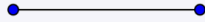
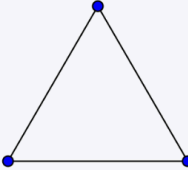
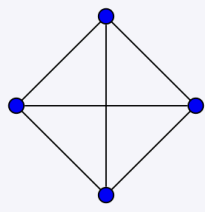
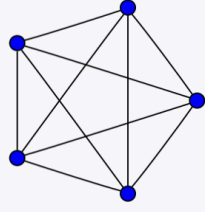
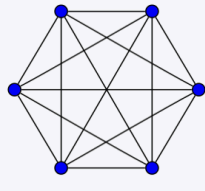
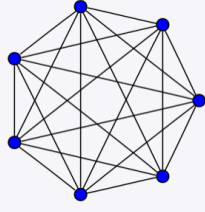
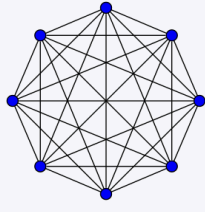
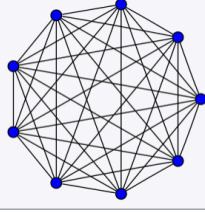
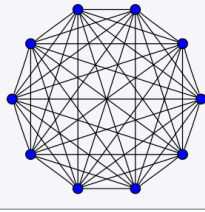
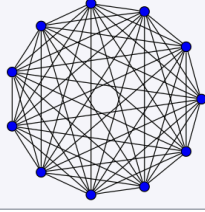
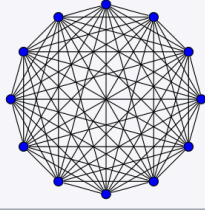
Cycle Graph:

- Cycles are simple graphs with $n \geq$ _____ vertices and edges: $\{1, 2\}$, $\{2, 3\}$, ..., $\{n-1, n\}$, and $\{ _, _ \}$.
- A cycle with n vertices is denoted as C_n .
- The total number of edges in C_n is _____.



Complete Graph:

- A simple graph of n vertices having exactly one edge between each pair of vertices is called a complete graph.
- A complete graph of n vertices is denoted by K_n .

$K_1: 0$	$K_2: 1$	$K_3: 3$	$K_4: 6$
			
$K_5: 10$	$K_6: 15$	$K_7: 21$	$K_8: 28$
			
$K_9: 36$	$K_{10}: 45$	$K_{11}: 55$	$K_{12}: 66$
			

- The total number of edges in K_n is _____ = _____ = _____.

Now, try to draw all possible simple graphs with 2 vertices.

Number of simple graphs with 2 vertices = _____.

Now, try to draw all possible simple graphs with 3 vertices.

Number of simple graphs with 3 vertices = _____.

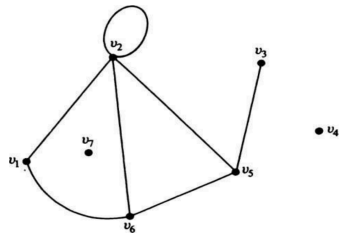
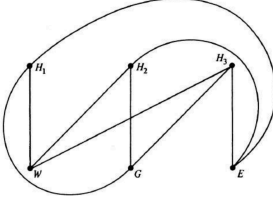
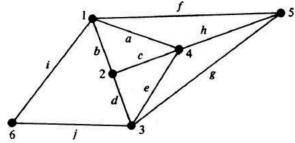
Now, try to draw all possible simple graphs with 4 vertices.

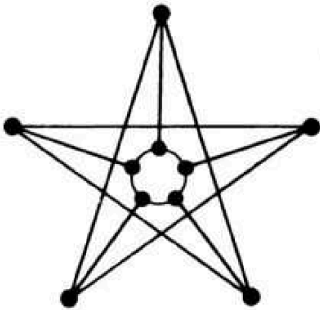
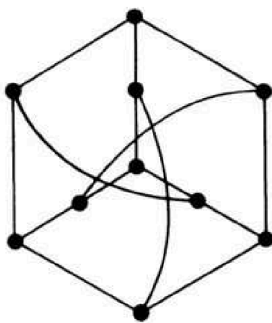
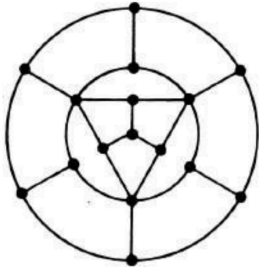
Number of simple graphs with 4 vertices = _____.

Observe the pattern. Think about the number of simple graphs with n vertices. 🤔

Hint: There are 2 choices (include and exclude) for every edge of a complete graph.

	$n = 3$	$n = 4$	n
Maximum degree of any vertex in a simple graph with n vertices			
Maximum number of edges in a simple graph with n vertices			
Maximum number of simple graphs possible with n vertices			

			
Number of Vertices			
Number of Edges			
Simple graph / Multi graph			
Number of pendant (degree-1) vertices			
Number of isolated (degree-0) vertices			
Is the graph complete?			
Is the graph regular?			
Degree sequence			

			
Number of Vertices			
Number of Edges			
Simple graph / Multi graph			
Number of pendant vertices			
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